Solutions

1.

Proof. We have $v_3 = v_1 + v_2$, so

$$v_1 + v_2 - v_3 = 0.$$

The coefficients $1,1,-1$ are not all zero. Hence the list is linearly dependent.

▫

2.

$$2 + 5x + 3x^2 \in \text{span}(p_1, p_2),$$

$$1 + x^2 \notin \text{span}(p_1, p_2).$$

Proof. For real scalars a and b,

$$ap_1 + bp_2 = a(1 + x) + b(x + x^2) = a + (a + b)x + bx^2.$$

Consequently, a polynomial $A + Bx + Cx^2$ belongs to this span if and only if there exist $a, b \in \mathbb{R}$ such that

$$a = A, \qquad a + b = B, \qquad b = C.$$

For $2 + 5x + 3x^2$, choosing $a = 2$ and $b = 3$ gives B =5.

For $1 + x^2$, suppose that $ap_1 + bp_2 = 1 + x^2$. Comparing the constant and quadratic coefficients gives $a = 1$ and $b = 1$, $a + b = 2$. Not 0. ▫

3.

Proof: Let

$$v_i = (x_i, y_i, z_i), \qquad i = 1,2,3,4.$$

Find scalars a_1, a_2, a_3, a_4, not all zero, such that

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0.$$

$$\begin{cases} x_1 a_1 + x_2 a_2 + x_3 a_3 + x_4 a_4 = 0, \\ y_1 a_1 + y_2 a_2 + y_3 a_3 + y_4 a_4 = 0, \\ z_1 a_1 + z_2 a_2 + z_3 a_3 + z_4 a_4 = 0. \end{cases}$$

There're three equations but four variables. After elimination, at least one variable is free. Therefore, a_1, a_2, a_3, a_4 could be not all zero. By definition, the four vectors are linearly dependent. ▫